EET202

HINES & TRANSFORMERS

MODULE: 4

e:1 – Transformer: Principle, truction and EMF Equation

Prepared By,

ADARSH SR
Asst. Professor
EEE Department
SCET Kodakara

Syllabus

ers –constructional details, principle of operation, EMF equation,
convention, magnetising current, transformation ratio, phasor
no load and on load, equivalent circuit, percentage and per unit
egulation. Transformer losses and efficiency, condition for
/A rating. Testing of transformers– polarity test, open circuit test,
npner's test – separation of losses, all day efficiency.Parallel
se transformers– numerical problems

Introduction

n static device

ectrical power from one circuit to another circuit
hout change in frequency.

y conversion

ks on the principle of mutual induction.

Introduction

dings wound on a common laminated core.

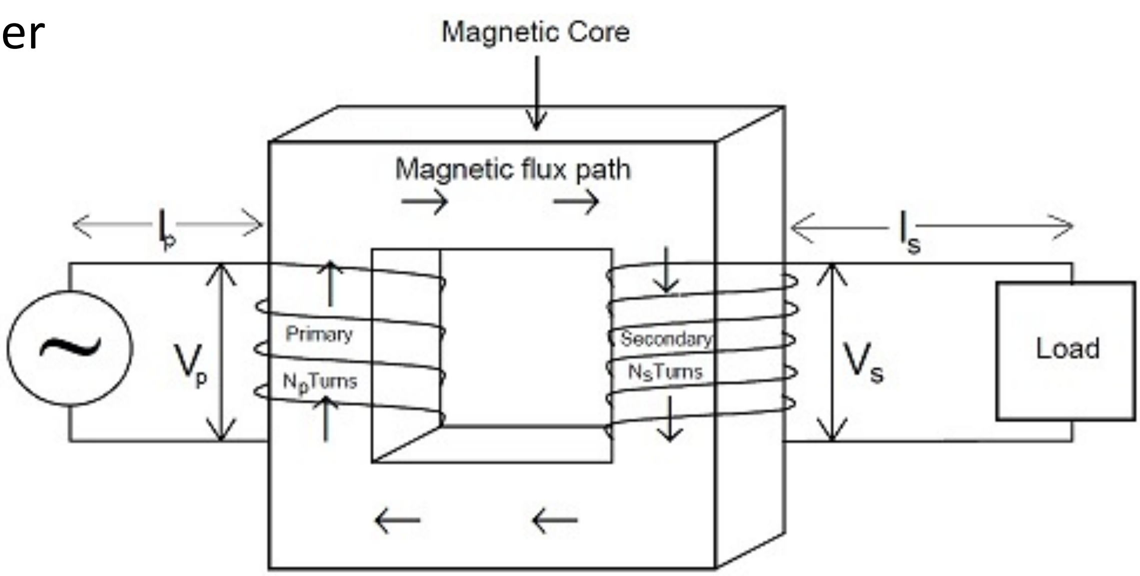
o the AC source is called primary winding.

n load is connected is called secondary winding.

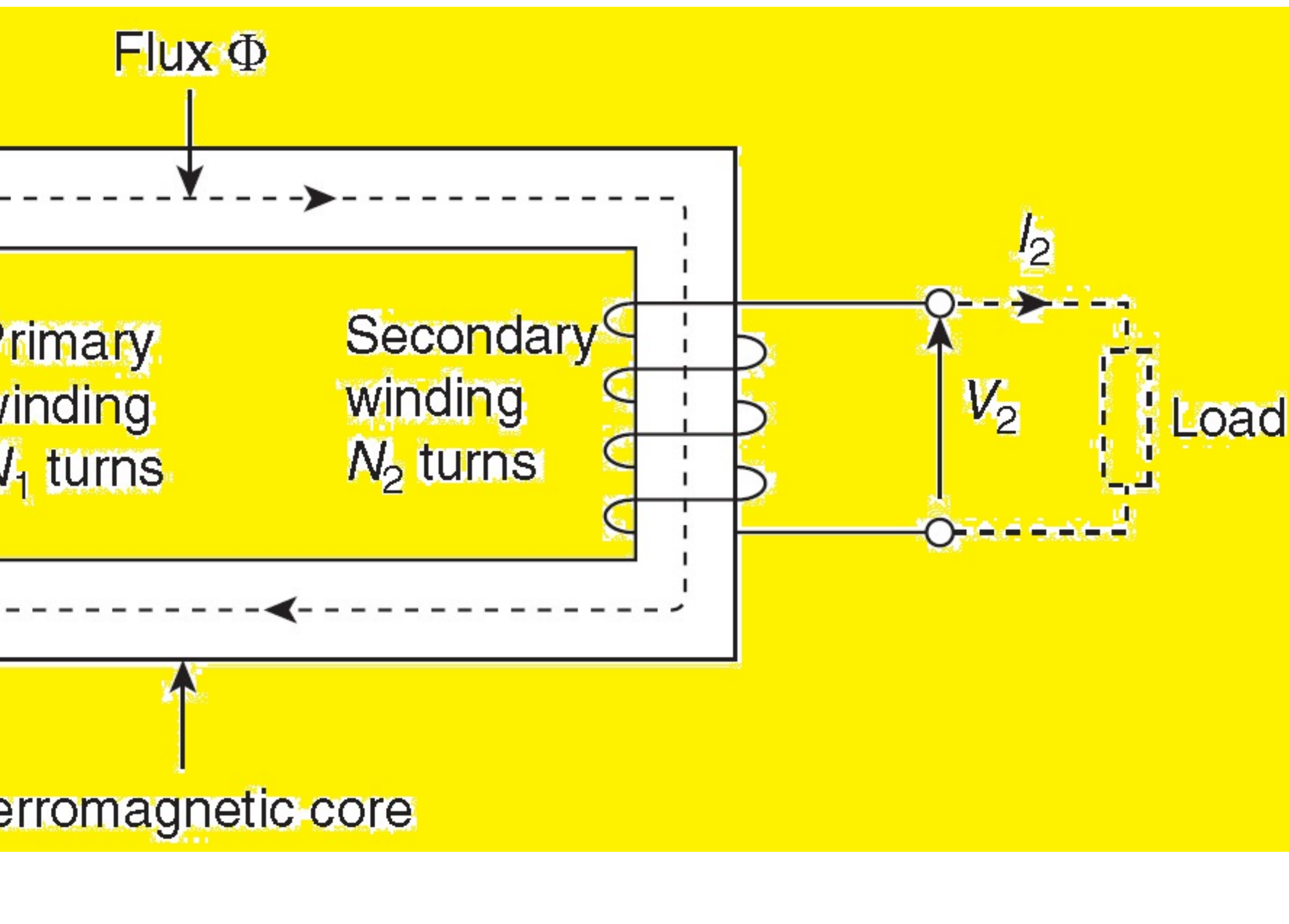
connection between two circuits.

ich the output voltage is greater than the input voltage is sformer.

ch the output voltage is less than the input voltage is called mer



ncipl e of Operation



nciple of Operation

rks based on the principle of mutual induction

ng voltage V_1 applied to the primary winding of an alternating current I_1 flows through it which ernating flux Φ in the core

raday's laws of electromagnetic induction, the duces emf in secondary winding (E_2).

Lenz's law induced emf opposes the cause

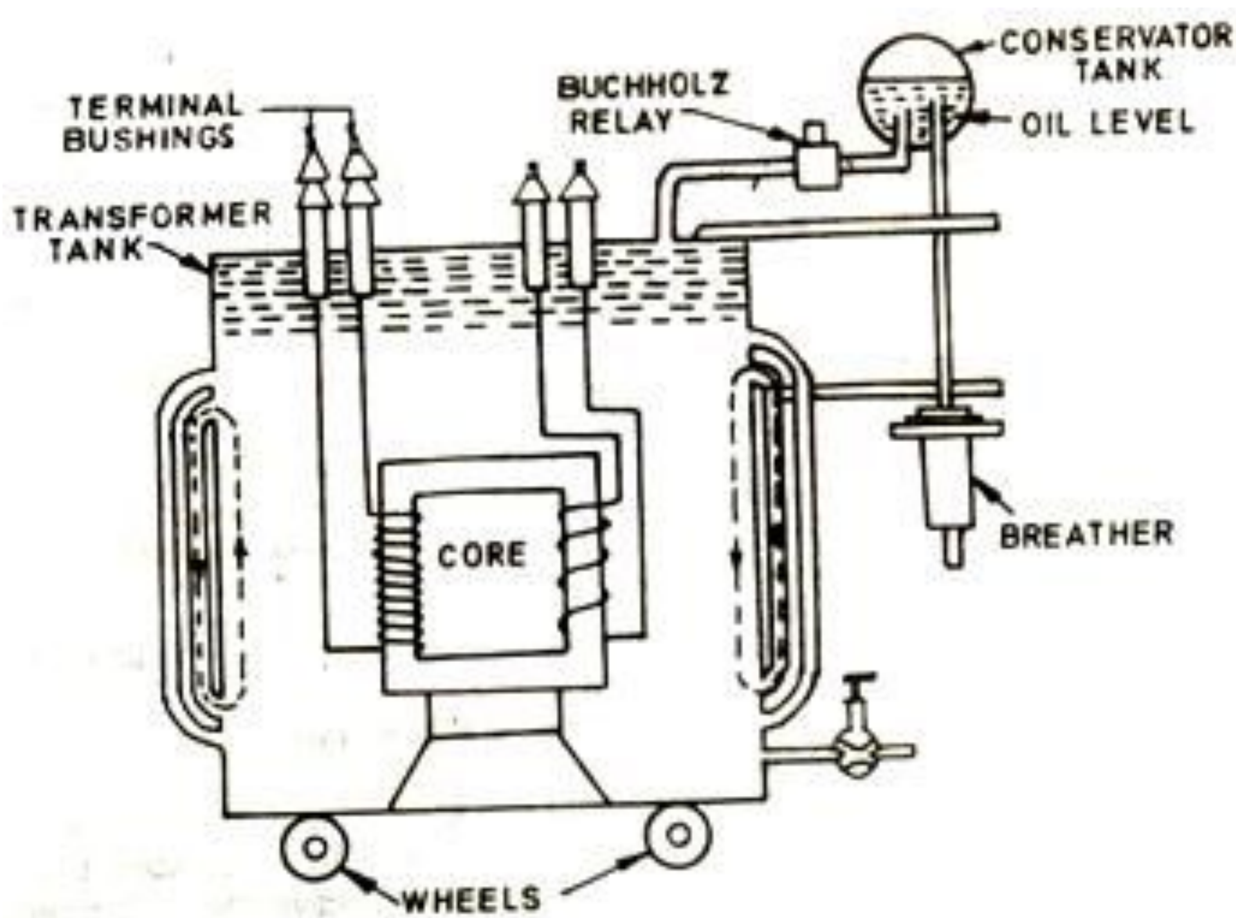
osed with a load, an output current I_2 flows d

Construction

ransformer are:

re:

ank



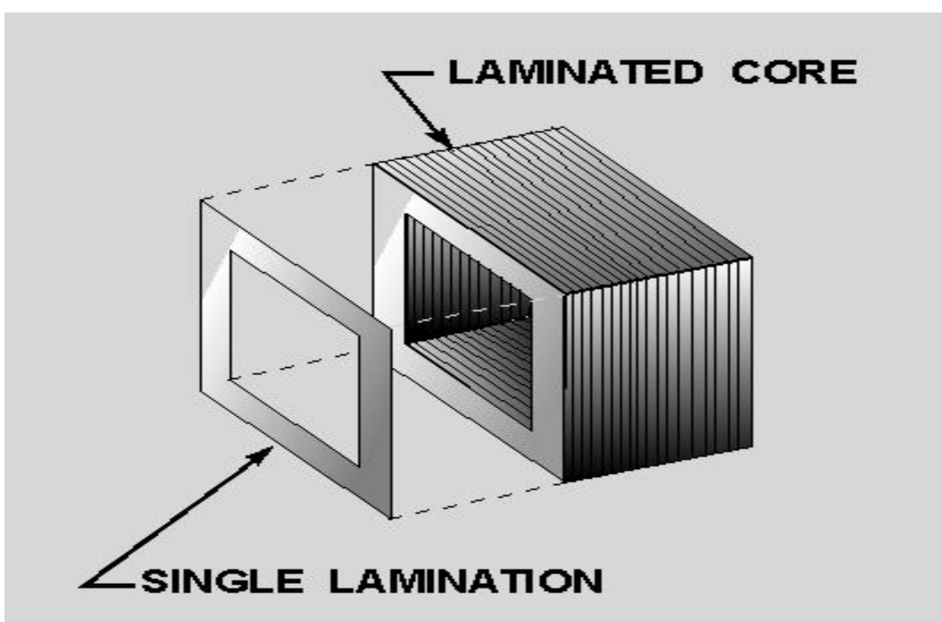
Construction

of laminations in order to minimize eddy current

on steel laminations (0.3 to 0.5 mm thick) are
hysteresis loss

ns are insulated from each other by using
rnish

re varnished

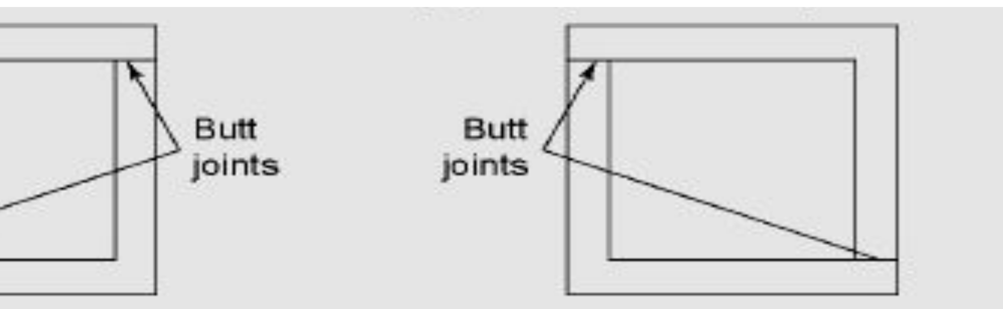


Construction

ed or E shaped laminations are used

h reluctance at the joint, the alternate layers are

ly to eliminate the joints

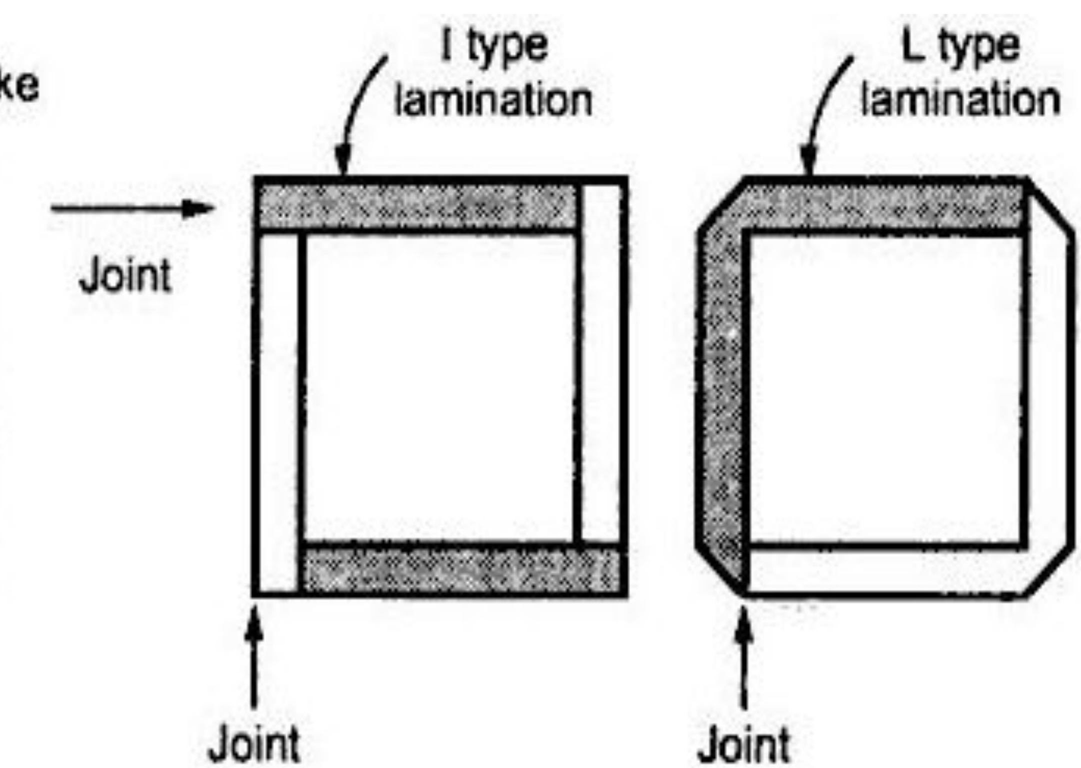


s for first, third,
of core type

(d) Laminations for second, fourth,
sixth layers of core type

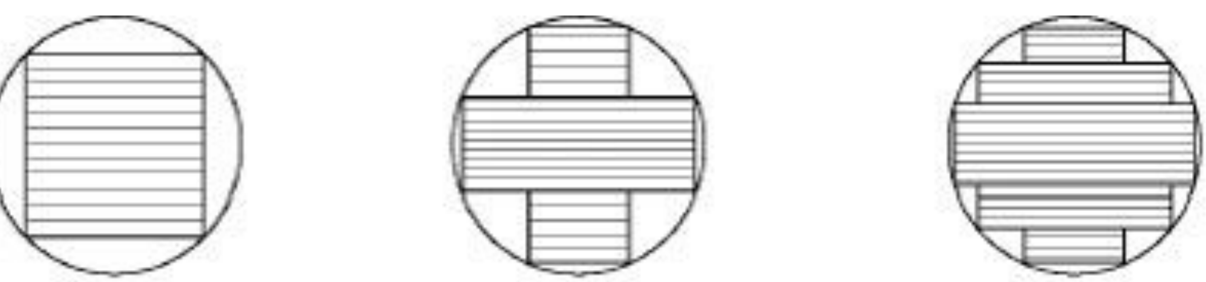
Construction

ion on which coils are wound is called limb while
n horizontal portion is called yoke of the core



Construction

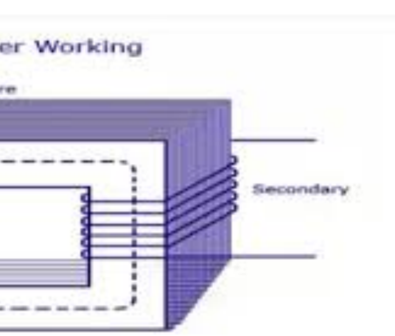
ss sections of limbs are shown in the figure

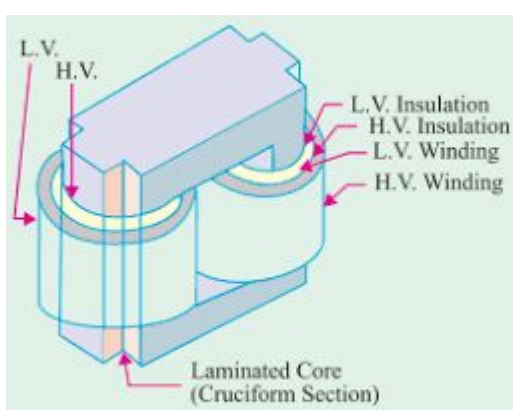


(b) Square

(c) Cruciform

(d) Three-stepped cruciform





Construction

a simple two-winding transformer consists of each winding eparate limb or core of the soft iron form which provides etic circuit.

rmer construction where the two windings are wound on ot very efficient since the primary and secondary windings from each other. This results in a low magnetic coupling ndings as well as large amounts of magnetic flux leakage.

ransformer can be improved by bringing the two windings with each other thereby **improving the magnetic coupling.**

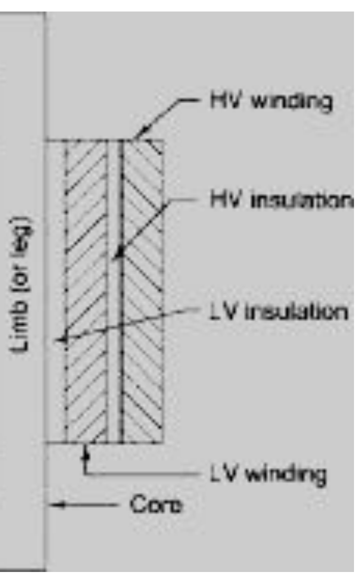
can be arranged as either concentric or sandwich.

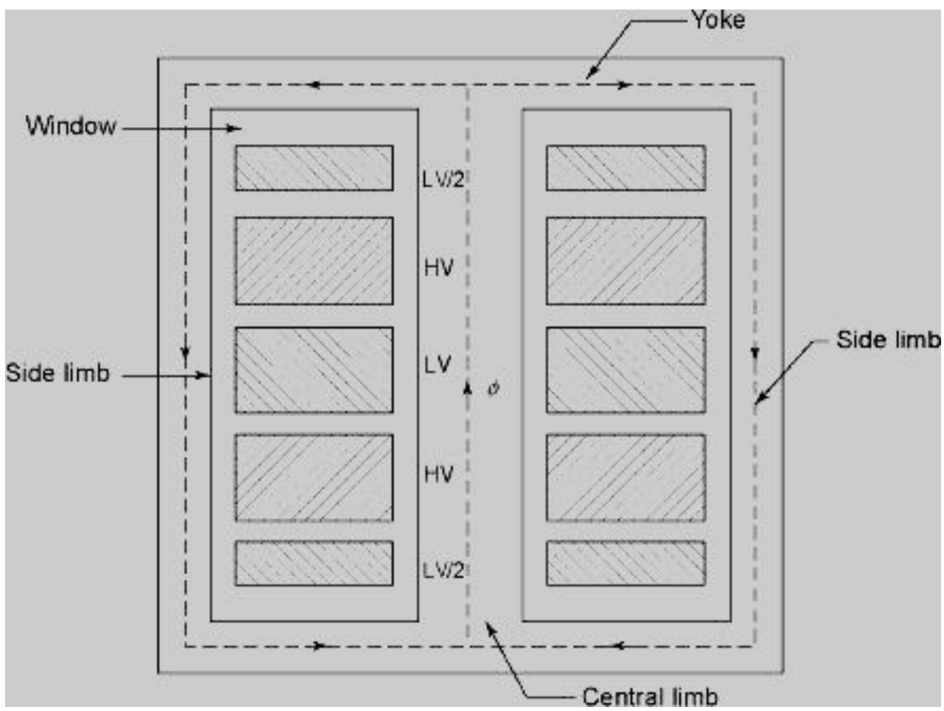
: HV and LV windings are arranged as concentric circles with

HV and LV windings alternate along the height of the limb.

Construction

ndwich Windings





Construction

nk

nk is completely filled with transformer oil.

core & windings are placed within a sheet
mersed in oil.

r oil provides insulation to the winding & core
he heat produced to the surrounding medium

nsformer oil
dditional insulation.
e insulation from dirt and moisture.
ay the heat generated in cores and coils.

Construction

ransformer changes with changes in temperature of oil ends upon the load on the transformer.

th increase in load and contracts when load decreases.

nds, the air is expelled out and air is drawn inside under This process is called **breathing**.

ccelerates the deterioration of oil due to increased rmation through moisture absorption and oxidation.

small auxiliary tank partly filled with oil and mounted rmer and connected to the main tank by a pipe.

Construction

s to keep the main tank of the transformer completely absorbing the oil during expansion or supply the oil with changes in temperature.

mpletely filled with oil but conservator tank is partially

Construction

pands, the air is expelled out and air is drawn inside
n of oil. This process is called breathing.

readily absorbed by oil and dielectric properties of

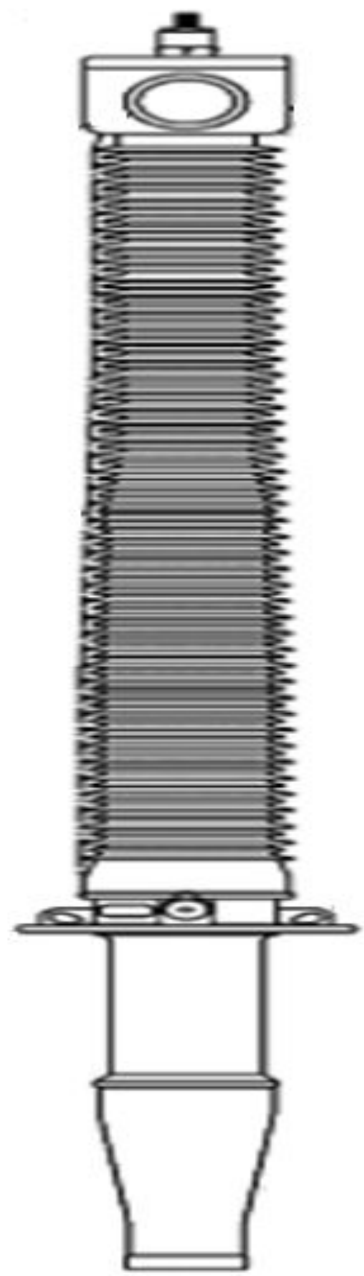
the transformer is made moisture free by letting it
apparatus called breather.

sts of a small container containing a dehydrating
a gel crystals impregnated with cobalt chloride. The
vhen dry and whitish pink when damp.

Construction

f the bushings is to insulate & to erminals of the transformer from ushings are made of porcelain

s of a current carrying element in conducting rod and a porcelain ved to isolate the current carrying



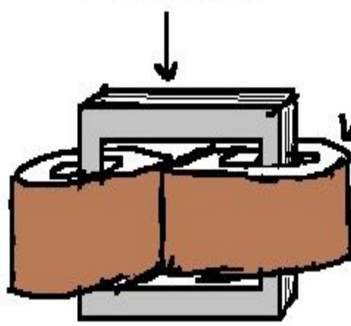
Construction

es of Transformers

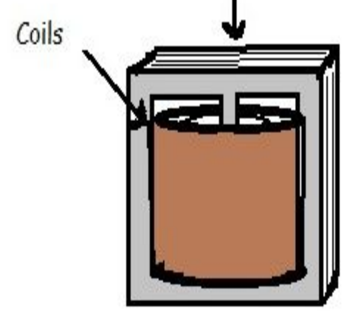
ction, Transformers are classified

sformer

sformer

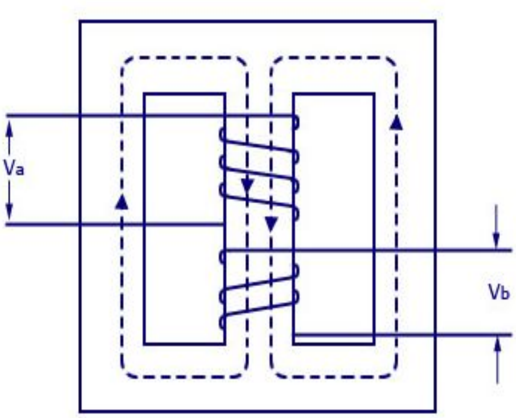


(a) Single Phase Core Type

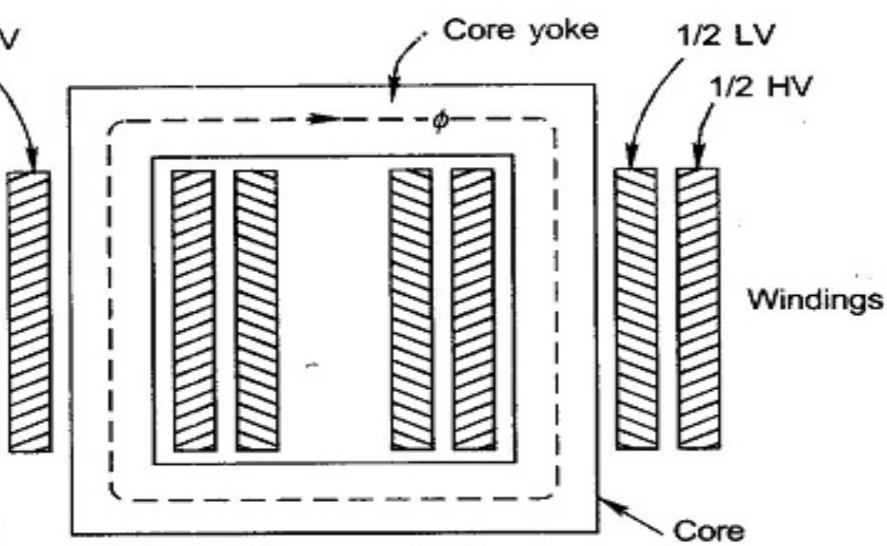


(b) Single Phase Shell Type

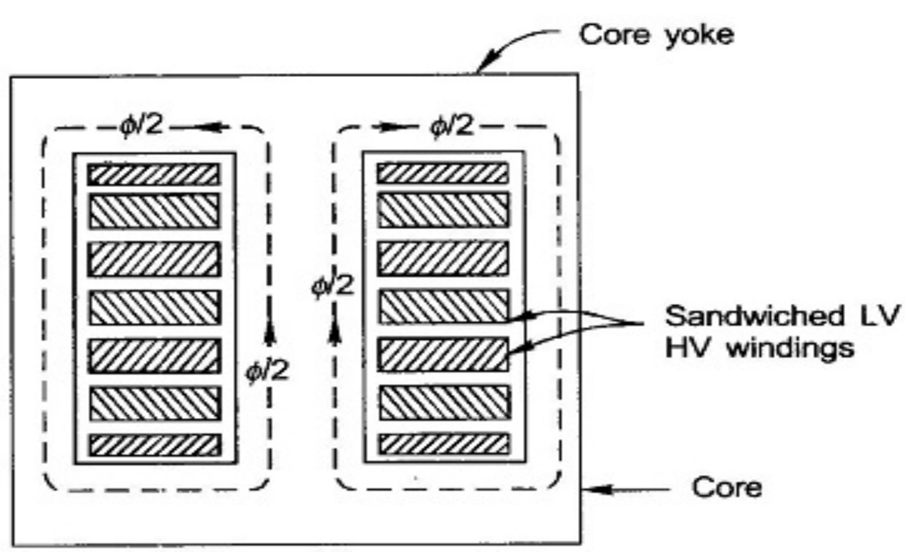
ype Transformer Winding



pes of Transformers



(a) Core-type transformer

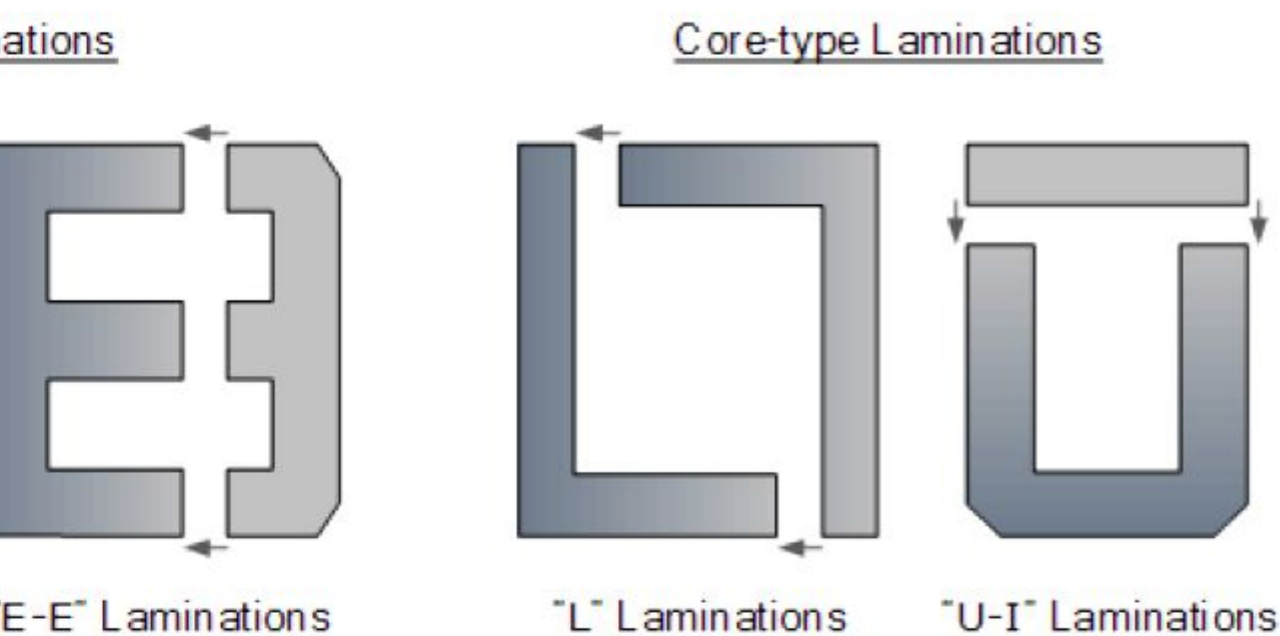


(b) Shell-type transformer

arison of Transformers

nsformer	Shell type Transformer
e core of the	Core surrounds the windings of the transformer.
wo limbs	The core have three limbs
magnetic circuit.	It consists of two magnetic circuits.
ing	It has sandwich winding
tively more	Poor natural cooling because the core surrounds the windings
ntle for repair	Dismantling for maintenance is relatively difficult
es are relatively	Core and copper losses are relatively low
ge and high	It is preferred for low voltage and low power applications

pes of Transformers



EMF Equation

l voltage of frequency 'f' is applied across the
it produces a sinusoidal flux of same frequency.

instant can be given by $\phi = \phi_m \sin \omega t$

day's Law,

$$e = -N \frac{d\phi}{dt}$$

F in Primary winding

$$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt}\left(\phi_m \sin \omega t\right)$$

$$= -N_1 \phi_m \omega \cos \omega t$$

$$= -N_1 \phi_m \omega \sin\left(90 - \omega t\right)$$

$$= N_1 \phi_m 2\pi f \sin\left(\omega t - 90\right)$$

EMF Equation

ge induced in primary winding

$$E_1 = \frac{N_1 \phi_m 2\pi f}{\sqrt{2}}$$

$$\boxed{E_1 = 4.44 N_1 \phi_m f}$$

ge induced in secondary winding

$$\boxed{E_2 = 4.44 N_2 \phi_m f}$$

N_1 : Number of primary turns
N_2 : Number of secondary turns
f : Frequency of AC supply
ϕ_m: Maximum value of Flux in Wb

nsformation Ratio

ondary voltage to Primary voltage is called tio of a transformer.

$$= \frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1}$$

ormer, K > 1

nsformer, K < 1

ner, Losses are zero.

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$